

TFE4120 Exams Aug 2024

Total 100 marks

Exercise 1: 10 marks (2, 4 and 4)

- A neutral atom is placed in an external electric field (E). What will happen? What is the atomic dipole moment (p) and how does it depend on the atomic polarizability (a)?
- The atom now is placed within the E at a random angle. Resolve and write the p in the two orthogonal directions.
- Consider an asymmetric molecule, write the complete polarizability tensor of the molecule when placed in an external E . This should consider all three dimensions.

Solution:

- The students should answer something like: even though it is a neutral atom, it comprises an electron cloud and nucleus of opposite charge. Therefore, when placed in an E the atom will be polarised. The enforced rearrangement of charges can be described by the p , where depends on a as: $p = aE$
- Similarly resolving p into the two orthogonal directions will be: $p = a_{\perp}E_{\perp} + a_{\parallel}E_{\parallel}$.
- This will include nine constants of a in x, y and z-axis.

$$\left. \begin{aligned} p_x &= a_{xx}E_x + a_{xy}E_y + a_{xz}E_z \\ p_y &= a_{yx}E_x + a_{yy}E_y + a_{yz}E_z \\ p_z &= a_{zx}E_x + a_{zy}E_y + a_{zz}E_z \end{aligned} \right\}$$

Exercise 2: 10 marks (2, 4 and 4)

- Write the equations for surface and volume bound charges.
- How you define p_f (free charges)? What is different from the bound charges?
- How can one prove the electric displacement (D) using the equation of the total charges (p_{total}) in the volume of a dielectric? What is the equivalent of Gauss law for the D ? Write both integral and differential form.

Solution:

- $\sigma_b = P \cdot \hat{n}$ and $p_b = -\nabla \cdot P$
- Free charges are what we add to the system (electrons/ions in conductors/dielectrics). Bound charges are fixed close to the atoms and usually 'stretch' the atoms under polarization, while free charges are moving around 'freely'.
- $p_{total} = p_b + p_f$, Gauss's Law: $\nabla \cdot E = \frac{p_{total}}{\epsilon_0}$ and $p_b = -\nabla \cdot P$

Then

$$\nabla \cdot (P + \epsilon_0 E) = p_f, \text{ and define as } D = P + \epsilon_0 E$$

$$\text{Gauss's law for } D, \text{ is } \nabla \cdot D = p_f$$

Or integral form: $\oint D \cdot da = Q_{f\text{enclosed}}$

Exercise 3: 20 marks (10 and 10)

A sphere of radius R is made of a material with dielectric constant of ϵ_r . In the sphere there is a uniform p_f .

- Find D inside and outside of the sphere.
- Using D find E inside and outside of the sphere.

Solution:

- We apply the Gauss law for D as follows.

$r < R$:

$$\left. \begin{aligned} \oint D \cdot da &= Q_{f\text{enclosed}} \\ Q_{f\text{enclosed}} &= p_f \frac{4}{3} \pi r^3 \end{aligned} \right\}$$

Therefore

$$D(4\pi r^2) = p_f \frac{4}{3} \pi r^3 \Rightarrow D = \frac{p_f r}{3}$$

$r > R$:

$$\left. \begin{aligned} \oint D \cdot da &= Q_{f\text{enclosed}} \\ Q_{f\text{enclosed}} &= p_f \frac{4}{3} \pi R^3 \end{aligned} \right\}$$

Therefore

$$D(4\pi r^2) = p_f \frac{4}{3} \pi R^3 \Rightarrow D = \frac{p_f R^3}{3r^2}$$

- Using D we find E in and out.

$r < R$:

$$D = \epsilon E = (1 + \chi_e) \epsilon_0 E = \epsilon_r \epsilon_0 E$$

$$E = \frac{p_f r}{3 \epsilon_r \epsilon_0}$$

$r > R$:

$$D = \epsilon_0 E + P, \text{ but } P = 0$$

Therefore

$$D = \epsilon_0 E \text{ and } E = \frac{p_f R^3}{3r^2 \epsilon_0}$$

Exercise 4: 30 Marks (10, 10 and 10)

A conductive sphere carries a charge q and has a radius R . The sphere is surrounded by a 'thick' metallic shell of inner radius a and outer of b . Shell and sphere are not in contact, and the shell has no charge. Consider infinity as a reference point in all cases.

- Make a sketch of the whole arrangement and indicate what are the charges in all surfaces. Find the surface charge density σ at R , a and b .
- Find the potential at the center of the sphere.
- You touch the outer side of the shell with a conductive wire that has its other end placed in the 'ground'. Explain what is going to happen and find the potential at the center of the sphere.

Solution:

- $\sigma = \frac{q}{4\pi R^2}$ at R , $\sigma = -\frac{q}{4\pi a^2}$ at a and $\sigma = \frac{q}{4\pi b^2}$ at b . The sketch should indicate all the q and $-q$ charges.
- As suggested, we consider infinity as a reference point and take into account potential contribution of all possible points.

In general:

$$V = - \int_o^r E \cdot dl, \text{ where } o \text{ is the reference point.}$$

In our case:

$$V = - \int_{\infty}^0 E \cdot dl = - \int_{\infty}^b \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr - \int_b^a 0 dr - \int_a^R \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr - \int_R^0 0 dr$$

$$V = - \int_{\infty}^b \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr - \int_a^R \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr$$

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{b} + \frac{q}{R} - \frac{q}{a} \right).$$

- The grounded wire drains the charge from the outer shell surface. Therefore, we should repeat the above but this time considering charges only on the two surfaces a and R .

$$V = - \int_a^R \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr$$

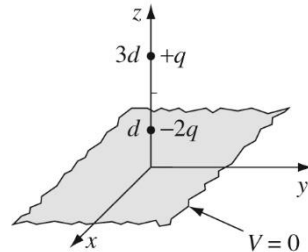
$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \frac{q}{a} \right).$$

Exercise 5: 10 Marks (2 and 8)

Considering the method of image charges, do the following based on the figure below:

- Draw the image charges and relevant distances.
- Find the force on the charge $+q$.

The grey plane is a grounded conductor. The x - y plane is placed is on $z=0$.

**Solution:**

- There are two image charges placed in $-3d$ and $-d$ with charges of $-q$ and $+2q$ respectively.
- Here we should consider the total distance between each of the charges

$$F = \frac{q}{4\pi\epsilon_0} \left[-\frac{2q}{(2d)^2} + \frac{2q}{(4d)^2} - \frac{q}{(6d)^2} \right]$$

$$F = \frac{1}{4\pi\epsilon_0} \left[-\frac{29q^2}{(72d^2)} \right]$$

Here we could add that all the charges and therefore the force are lying on z -axis.

Exercise 6: 20 marks (10 and 10)

Prove the following Maxwell's equations:

- $\nabla \times B = \mu_0 J$ considering Ampere's Law, think how to relate total current enclosed to the volume current density J and then apply Stoke's theorem.
- $\nabla \cdot E = \frac{\rho}{\epsilon_0}$, considering Gauss's Law and divergence theorem then think how to relate the total charge enclosed to the charge density ρ .

Solution:

- For $\nabla \times B = \mu_0 J$ proof briefly:

Ampere Law in integral form is given as $\oint B \cdot dl = \mu_0 I_{enc}$,

Where the enclosed current is $I_{enc} = \int J \cdot da$

Applying the Stokes' theorem to the closed integral $\oint B \cdot dl = \mu_0 I_{enc} = \mu_0 \int J \cdot da$

$$\int \nabla \times B \cdot da = \mu_0 \int J \cdot da$$

Dropping the same integrals from both sides results in:

$$\nabla \times B = \mu_0 J$$

b) For $\nabla \cdot E = p/\epsilon_o$ proof briefly:

Gauss's Law in the integral form is given as $\oint E \cdot da = \frac{Q_{enc}}{\epsilon_o}$.

Applying the divergence theorem for E relates a closed surface to a volume integral:

$$\oint E \cdot da = \int (\nabla \cdot E) d\tau$$

Now Q_{enc} is relates to volume integral as follows:

$$Q_{enc} = \int p d\tau$$

Then the above divergence theorem becomes:

$$\int (\nabla \cdot E) d\tau = \int p/\epsilon_o d\tau$$

Dropping the volume integrals from both sides results in:

$$\nabla \cdot E = p/\epsilon_o$$